

StudioStack: An Inventory Management System

Project Documentation

Team Name:

StudioStack

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CIS 4914 Senior Project

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Final Presentation Video:

<https://youtu.be/-KZjjempejU>

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Keywords

Inventory Management, User Experience, Custom Software Solutions, Reservation System,
Full-Stack Development, Firebase, ReactJS, Barcode Integration, Web Application, Equipment
Checkout System, Software Design Engineering, Data Management Systems

Project Overview

Project Name and Description

StudioStack is an inventory management web application that aims to provide the Production Department at The Agency at UF with a smooth and effective way of reserving, checking out, and managing equipment. The features that were implemented in this web application include but are not limited to: scanning a barcode to check out a piece of equipment, choosing the date range and specific equipment for a reservation, and being able to report broken equipment. It also has admin features such as adding and removing items from the inventory, updating users, and reviewing various statistics. This system will increase the efficiency and productivity of the production department by reducing the time taken to check out items, increasing accuracy, and improving the overall user experience for students and department managers. StudioStack is tailored to meet the needs of The Agency at UF while being cost-effective and customizable for the future growth of the production department.

Project Goals and Objectives

The goal of this project was to pinpoint an issue that The Agency was having and find a way to solve the problem using software and our skill sets. Prior to working on this project, the Production Department had been struggling to coordinate equipment reservations in the department, as they were using an Excel spreadsheet, which was inefficient and time consuming. On a broader scale, the main objective of this project was to facilitate an easier experience for the production department by creating an inventory management system called StudioStack, which would be inexpensive and tailored exactly to their needs. Our main goals for StudioStack were to

create a page for users to create reservations ahead of time and be able to check in/out an item when they are ready, report an item with an issue, keep track of the department's inventory, and allow admins to view statistics on equipment.

Project Scope

StudioStack was designed to have eight pages: Home/Dashboard, Calendar, Reservations, Reports, Statistics, Teams, Users, and Inventory. Users have access to each page depending on whether they are designated as admin or student users. The core functionality for this system included scheduling reservations, reporting items, and viewing statistics. The website was developed over a period of 4 months (one academic semester), which included proposing the project, creating the wireframe/user interface design, implementing the code, testing/debugging, deploying the website, and creating documentation for the project. We utilized React, Node.js, Tailwind CSS, and Firebase in the development of StudioStack. There were little to no costs associated with this project. Our website uses a scanner to check in/out equipment, which was purchased with the permission of our advisor. We also had to upgrade to Firebase's Pay as You Go (Blaze) plan to use Firebase's email trigger functionality. However, our project is expected to operate under the free tier of this plan, unless the number of emails exceeds a certain amount per month.

Team Members: Roles and Responsibilities

Ashley Guerra:

- Roles: Project Manager, Full-Stack Developer
- Responsibilities:

- Navbar
- Inventory Page
- Adding & Removing Items from Inventory
- Downloading an item's barcode
- Filter inventory items by category
- Reports Page
- Submitting Reports
- Reviewing and Resolving Reports
- Admin & User Dashboard
- Notifications
- Teams Page

Maeson Vallee:

- Roles: Scrum Master, Full-Stack Developer
- Responsibilities:
 - Authentication
 - Student vs Admin
 - Users Page
 - Adding & Removing Users
 - Reservations Page
 - Creating Reservations
 - Checking Out & In Items
 - Calendar Page

- Statistics Page
- Overdue Item Email

Deliverables

Completed Deliverables and Functional Specifications

Completed deliverables include:

- Login Authentication:
 - The program checks if the user is an approved user in the database.
- Dashboard:
 - Student: The user can quickly access their reservations and view notifications.
 - Admin: The user has quick access to the user, inventory, teams, and reports page.
They can see notifications and their own reservations.
- Calendar:
 - Users can click on reservations in the calendar and look at the information about it. The information displayed is dependent on whether the user is a student or an admin.
 - Users can go forwards and backwards in time to see all the reservations made.
- Reservations:
 - Creating Reservations:
 - Users can create reservations, filling out the reservation name, team/client name, check out date, check in date, and which specific items they want to check out.
 - Checking In/Out Items:
 - Users can scan a barcode to check out/in the items in their reservation.
 - They can also extend the reservation, cancel the reservation, and make

changes to the items (add/remove) in the reservation.

- Reports:
 - Users can create reports for faulty items, which alerts the admins via email notification.
 - Admins can resolve reports submitted by users. Reported items are not allowed to be checked out for a reservation. Until the issue is resolved, they are temporarily unavailable.
- Statistics:
 - Only admins can see the statistics page.
 - Admins can see what items are overdue, the record of users who have had overdue items, and various other statistics regarding equipment, reports, and users.
- Users:
 - Only admins can see this page.
 - Admins can add users, remove users, and change the roles of the users.
- Inventory:
 - Users can view a list of items/equipment in the production department. They can see the item's category, status, and download barcodes.
 - Admins can add or remove items from the inventory.
 - Users can search and filter through the inventory.
- Teams/Clients:
 - Users can see a page with all the internal and client teams being worked on in the production department.

- Admins have the ability to add or remove a client or internal team.

More Functional Specifications:

Once an item is overdue, an email automatically gets sent to all the admins and the user who has the overdue item. This email states the reservation and which item is overdue. Also, if an item gets reported and it is later used in another reservation, an email gets sent to the user who made that later reservation and allows them to change the equipment in the reservation to a non-broken one.

Once a user submits a report for an item, StudioStack admins will receive an email with further details about the report, which can also be viewed in the Reports page. Admins receive a notification on their dashboard when a report is sent, and users are sent a notification to their dashboard when their report has been resolved by an admin.

Current Project Status

The website itself is functional and done, except for the few remaining work that is listed below. The main branch in the GitHub repository is representative of all the work that we have done this semester. This project is still usable without implementing the small remaining work list below. However, the Production Department closet should be set up before this website fully gets used.

Remaining Work and Backlog Items

Remaining work includes:

- Sending an email reminder 12-24 hours before a reservation is due
- Archive reservations and reports that are over a month old.

- Being able to change the reservation name, team, etc. after the fact.
- Predict finance expenses: if a piece of equipment seems to be breaking after a set amount of time (ex: 3 months), keep track of having to purchase a new one and the prices of it.
- Add key/legend to the Inventory page for item statuses.

The office setup for the checking out and in of items needs to be completed as well. This includes putting a desktop in the Production Department closet, connecting the scanner to it, making a binder of barcodes, and labelling each individual equipment with the barcodes.

Technical Details

Codebase Overview

In our project directory, there are folders *.github/workflows*, *functions*, and *src*. Inside the *.github/workflows* folder, it houses a file that is used to deploy the firebase hosting each time main is updated. Inside the *functions* folder is where the cloud functions are hosted, which is inside *index.js*. Finally, the *src* folder houses the bulk of our code. Inside this folder is an *assets* folder, which contains the png images used in the website. The *components* folder has the files for all the components used, such as labels, pop ups, notifications, and the header. The *firebase* folder contains the firebase configuration file. The *pages* folder contains all of the pages for the website, such as the Dashboard, Calendar, Statistics, etc. Finally the last of the important files in the *src* folder are the *App.jsx* and *main.jsx*.

Key Dependencies and Libraries

Our project uses ReactJS, a library used for building UIs for web applications. Our key dependencies include fullcalendar, tailwindCSS, bwip-js, hamburger-react, react-dom, react-select, react-calendar, react-router-dom, reactjs-popup, and recharts. It also includes firebase dependencies such as firebase-admin and firebase-functions.

Deployment Instructions

To run the code locally, in the project directory, run:

```
npm i
```

```
npm run dev
```

In the GitHub Repository, everytime there is a PR into main, a workflow to automatically deploy the hosting will run. To manually deploy the hosting, in the project directory, run:

```
firebase deploy --only hosting
```

To deploy the cloud functions, in the project directory, run:

```
firebase deploy --only functions
```

Database Schema and Data Migration Details

Database schema:

- Inventory:
 - id: string, ex: “0hyH3DZcaYKjTvaPCn4c”
 - availability: string, ex: “available”
 - category: string, ex: “Camera & Accessories”
 - name: string, “Atomos Ninja V Monitor”
 - timestamp: TIMESTAMP
- Mail:
 - id: string, ex: “0IZpYiGVbdKYCEE0qkTV”
 - Delivery: not important in regards to our data/application.
 - message:
 - html: string, ex: “ <p>Item Name & ID: Selens 5-in-1 Diffuser & Reflector (3mYUHA9jcLPI6Kk46LSC)</p> <p>Type: Extreme

Heat/Cold Damage</p> <p>Reported By:

ashleyguerra@ufl.edu</p> <p>Description: Test</p> “

- subject: string, ex: “StudioStack Item Report”
- text: string, ex: “There is an issue with an item.”
- to: array, ex: [“theagencyatufdevs@gmail.com”]
- type:
 - subject: string, ex: “Extreme Heat/Cold Damage”
- Notifications:
 - id: string, ex: “7GyBRIVvQsbmTiGZkN3r”
 - adminClosed: bool
 - amount: string, ex: “3”
 - reservationName: string, ex: “Test Notif”
 - time: TIMESTAMP
 - type: string, ex: “overdue”
 - userClosed: bool
 - userEmail: string, ex: “valleemaeson@ufl.edu”
- Reports:
 - id: string, ex: “0cWl3gIemb8moHuDFV9u”
 - item: string, ex: “Boom Mic Clip”
 - itemId: string, ex: “NGmmSWgi5kXPUR9TQeCX”
 - message: string, ex: “Test Report for User Manual”
 - resolved: bool
 - resolvedBy: string, ex: “valleemaeson@ufl.edu”

- resolvedOn: TIMESTAMP
- subject: string, ex: "Item Cannot Connect to App"
- timestamp: TIMESTAMP
- user: string, ex: "valleemaeson@ufl.edu"
- Reservations:
 - id: string, ex: "1xOj3j1wzTw3TdZ43rLv"
 - checkedOutItems: array, ex: [id: "xwztVouQmjgk3IfkGO23", name: "50mm EF mount lens"]
 - checkedInItems: array, ex: [id: "xwztVouQmjgk3IfkGO23", name: "50mm EF mount lens"]
 - endDate: TIMESTAMP
 - equipmentIDs: array, ex: [id: "xwztVouQmjgk3IfkGO23", name: "50mm EF mount lens"]
 - name: string, "example reservation 2"
 - overdueItems: array, ex: [id: "xwztVouQmjgk3IfkGO23", name: "50mm EF mount lens", time: TIMESTAMP]
 - startDate: TIMESTAMP
 - team: string, ex: "Bartram"
 - userEmail: string, ex: "valleemaeson@ufl.edu"
 - verifiedBy: TIMESTAMP
- Teams:
 - id: string, ex: "Bartram"
 - type: string, ex: "Client"

- Users:
 - id: string, ex: “valleemaeson@ufl.edu”
 - isAdmin: bool

If this project is to be moved away from Firebase, you can export the data from the firestore and use a script to transform it into a format that is compatible with the target database system. After transformation, the data can be inserted into the new database using appropriate client libraries or bulk import tools.

Software Architecture

A user interacts with the front end of our web application. Our front end consists of ReactJS and TailwindCSS. We have third party applications/code that interact with the front end, such as Google Authentication Service, Trigger Email Extension, and Barcode Generator Library. Our front end also interacts with our back end. Our backend consists of a web server that interacts with our file system and our database, which is the Firestore database. A model of our Software Architecture is shown in Figure 1 of the Appendices.

Testing Information

Summary of Testing Conducted

We tested our web application by having members of the production department directly interact with all the features on StudioStack. We invited the Production Department Manager and a few lead members to The Agency office, where they created their accounts upon arrival. We assisted members of the team with creating accounts for the first time, adding users, and providing admin permissions. Then, we walked through the two scenarios that the users would perform on the website for testing purposes.

Test Cases and Results

Our first external test case was for the Reservations page implementation. After the members logged into their own accounts, we asked them to create a reservation. They were tasked with entering a name, selecting a team name, a check-out and check-in date, and the items they wanted to reserve. We received expected results: the user was able to successfully create a reservation with all the proper information.

Our second external test case was for the Reports page functionality. We asked users to create a report, view a report, and then asked the admins to resolve it. Users were directed to the Reports Page on StudioStack, where they submit a report with information about an item, such as the type of issue, the item name, the item ID, the user reporting it, and a brief description. While the report went through and the email was sent, one user unexpectedly received an alert that their report was missing information - a “catch” error that indicates a field was left blank when the report was submitted.

We performed additional testing internally as we developed StudioStack. We each tested as much as possible for edge cases on the pages we worked on, and most, if not all, important features utilize try/catch blocks to ensure the code is working as expected. In the case of a failure, the user is alerted of what the issue is.

Page/Feature	Test Case Description	Steps	Expected Result	Actual Result	Status
Reservations	Creating a Reservation	1. Open the Reservations Page 2. Click on the Teal plus icon on the top right 3. Enter a reservation name, select a team, select a check out and check in date, and select item(s) to reserve.	A reservation is successfully created. The user can see it under "Active Reservations".	A reservation is successfully created. The user can see it under "Active Reservations".	Completed
Inventory	Adding an item to the inventory	1. Open the Inventory Page 2. Click on the teal plus button on the top right 3. Enter the name of an item 4. Select a category 5. Click submit	The item is added to the top of the inventory	The item is added to the top of the inventory	Completed
Users	Adding a user without a UFL email	1. Open the Users page 2. Click on the teal plus button on the top right 3. Enter an email with a domain that is not "ufl" 4. Click submit	The user receives an alert indicating that the user needs to be a UF student, so only a UFL email may be used	An alert pops up on the top of the page that states, "Invalid email. Must be a UFL email and not already in the database."	Completed
Teams	Adding a Team that already exists	1. Open Teams page 2. Click on the teal plus icon on the top-right of the page	A duplicate team name should not show up in the list.	A duplicate team name does not show up in the list.	Completed

		3. Enter a team name that is already listed			
--	--	---	--	--	--

Outstanding Bugs and Defect Log

Page/Feature	Description	Status	Comments
Report	False alert pops up when a user submits a report indicating that a field has been left blank, but the report still goes through and email is sent.	Unresolved	Upon further inspection of the faulty submitted report, its associated document in Firebase, and repeated testing, we have not been able to resolve the issue. The false alert remains an outstanding bug, although the reporting functionality still works.
Reservation and Report Label	On some computers, the arrow button is centered, but on others it is too low.	Unresolved	We think it has something to do with the transform styling.
Notification Label	On some computers, the close button is centered, but on others it is too high.	Unresolved	We think it has something to do with the transform styling.
Hover Styling	When using Tailwind CSS, the 'hover:' is not working on the hosted web app. It is also not working for some localhosts. Even when hovered with a mouse, the styling does not change.	Unresolved	When researching this problem, it seems that it has been happening for other users. It is connected to a Tailwind V4 issue.

User Documentation

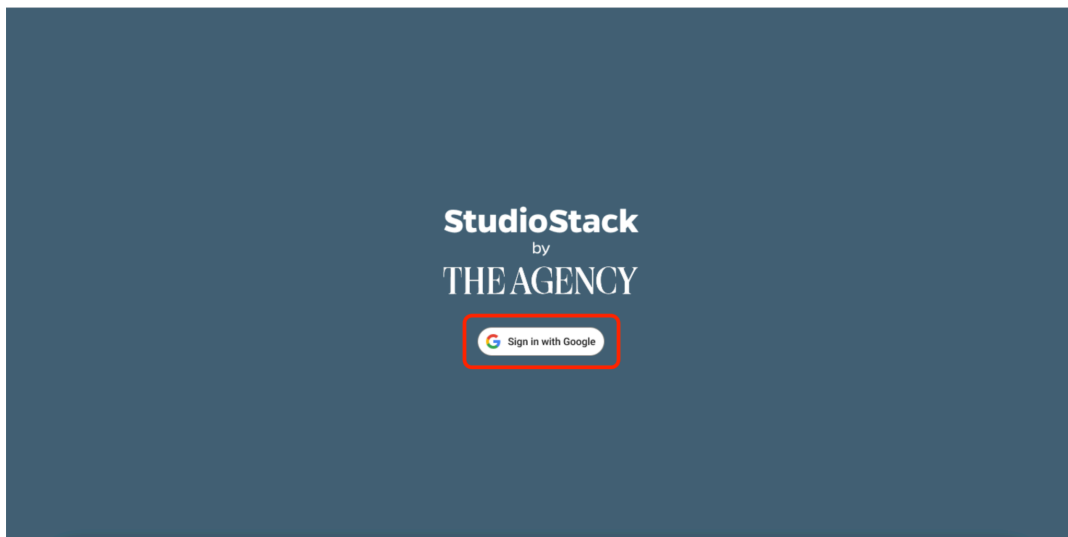
StudioStack User Guide

I. Introduction to StudioStack

StudioStack is an inventory management system that aims to provide the production department at The Agency at UF with an effective way of checking out equipment. The features of this web application include but are not limited to scanning a barcode to check out a piece of equipment, choosing the date range for the reservation, being able to report broken equipment, and having admin features such as adding and removing items from the inventory. StudioStack is tailored to meet the needs of the Agency at UF while being cost-effective and customizable for the future growth of the production department.

II. Logging In

To log in, click on the “Sign in with Google” button (circled in red) and choose which account you would like to log in with. This account should be your ufl email address.



If you experience any issues logging in, please contact your department manager to ensure you have the correct permissions.

III. Dashboard

The dashboard / home page allows all users to see their upcoming scheduled reservations and reserved equipment. Users can click on their reservations, which takes them directly to the check-out/check-in page for that reservation. Users are also able to see what items

they have reserved for check-out under all active reservations. Both types of users have a notification center for different purposes.

Upcoming Reservations

Testing scroll
03/19/2025, 05:12:00 PM - 04/05/2025, 05:12:00 PM

User Manual
03/31/2025, 11:14:00 PM - 03/31/2025, 11:20:00 PM

Reserved Equipment

Item	ID
Sand Bag (Large)	9bekRiZUdg7rPHx33ONA
Canon C200	43x5KqAcPHTxNMxE0vye
Display Turntable	A24SaYNbYY8wnFeN7g1N

Student

The student's dashboard features a mini, at a glance calendar that they can look through and a notifications center. Notifications will pop up here once an admin has reviewed and resolved a report that the user submitted. They will also receive notifications on the dashboard and via email when they have any overdue items for a reservation. Users can click on the 'x' to remove a notification from their dashboard. The student's navigation bar features four tabs: Home, Calendar, Reservations, and Reports.

StudioStack

HomeCalendarReservationsReports

Welcome, Student!

« < March 2025 > »

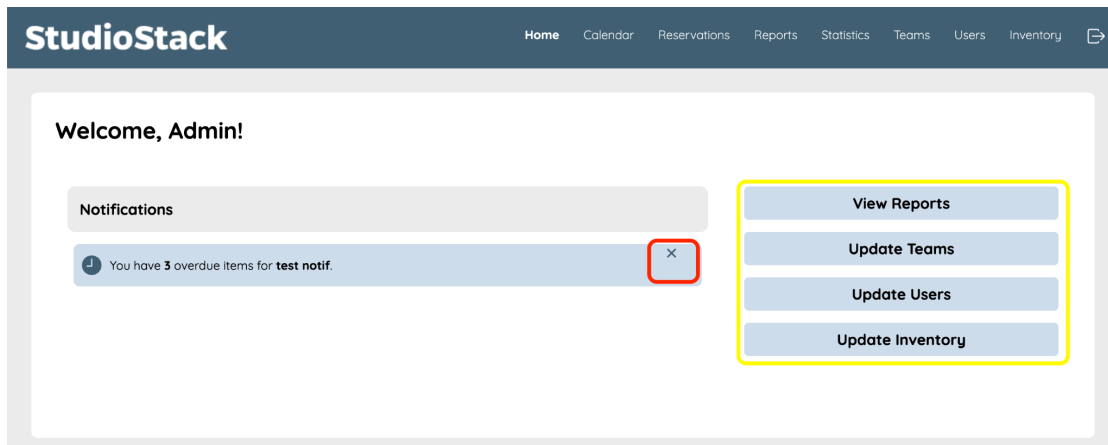
MON	TUE	WED	THU	FRI	SAT	SUN
24	25	26	27	28	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

Notifications

A report for Boom Mic Clip was resolved by ashleyguerra@ufl.edu on 03/31/2025, 10:03:32 PM.

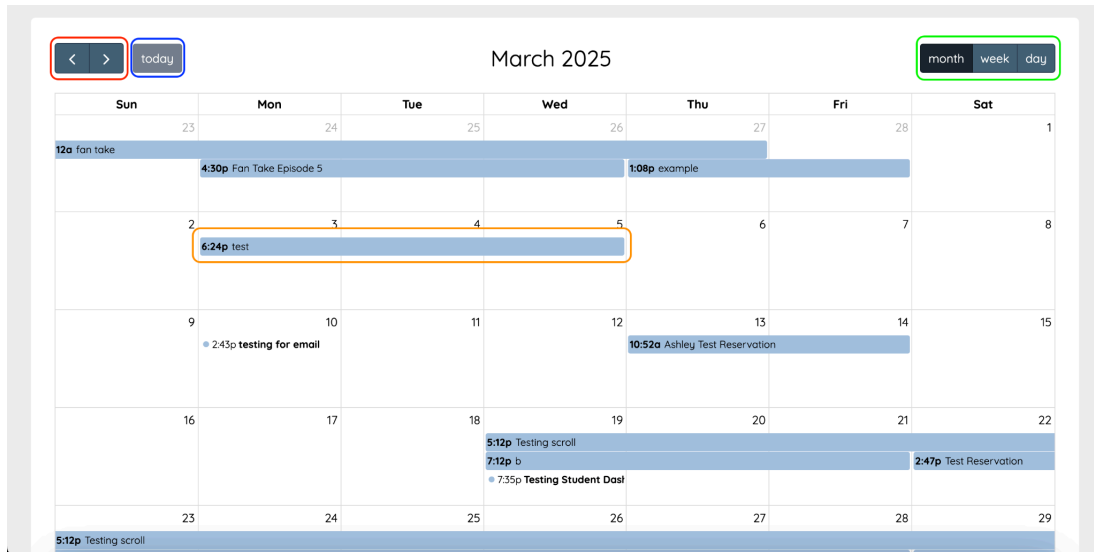
Admin

The dashboard for admins features four buttons to view reports, update the teams (client or internal), update the users, and update the inventory. The admin dashboard also has a notification center, which displays notifications when a user has submitted a report that the admin needs to review or when a user has items that are overdue. Admins can remove both types of notifications by clicking on the 'x'. Report notifications also automatically disappear once the admin has resolved the report associated with that notification. The admin's navigation bar includes seven tabs: Home, Calendar, Reservations, Reports, Statistics, Users, and Inventory.



IV. Calendar

This calendar page allows all users to see what reservations have been made and when. It allows them to change the calendar view to see reservations in the past and the future. You can do this by clicking on the forward and backward arrows in the red box. If you click on the "today" button in the blue box, it brings you back to the current day. You can click on the green box to change the calendar view to month, week, or day. To see a specific reservation, click on one in the calendar, as shown below in the orange box. After clicking, the page will automatically scroll down to the information.



Student

After clicking a certain reservation, students can see certain stats about a reservation on the bottom of the page. An example is shown below:

sample reservation

Team: Disney

3/29/2025, 7:25:00 PM - 4/1/2025, 7:25:00 PM

Equipment Reserved: Atomos Ninja V Monitor
DJI Ronin RSC2 Camera Gimbal

Admin

After clicking a certain reservation, admins can see certain stats about a reservation on the bottom of the page. An example is shown below:

ex

Team: Disney

3/27/2025, 2:43:00 AM - 3/31/2025, 2:43:00 AM

Reserved by: valleemaeson@ufl.edu

Equipment Reserved:

Item	ID	Status
SD Card - 128GB	9F1znxjL7Rym7Fr5D74Y	Awaiting Checkout
SD Card - 128GB	Bf1lmzJqfn43nXoYMNCf	Awaiting Checkout
Canon EFS 18-135mm Macro Lens	BsAoLQLtSkyqqa6en4Vu	Awaiting Checkout

V. Reservations

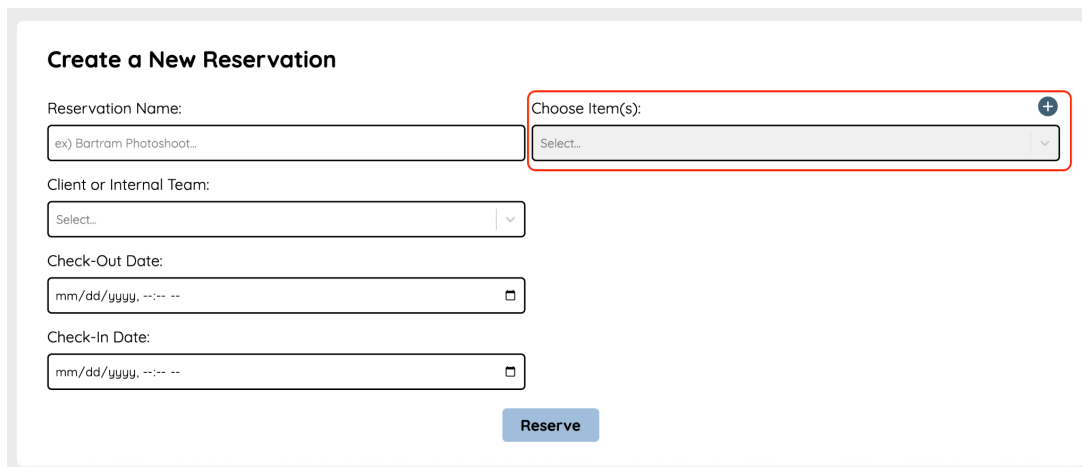
The reservations page shows all reservations made by the user. It is split into active and past reservations. You are also able to see any reservation on the calendar page. To view a specific reservation that you made click on the arrow inside the reservation label, as shown in the red box.



The screenshot displays the 'Reservations' page. At the top right is a plus button (+). Below this, the 'Active Reservations' section contains a reservation card for 'photoshoot' with 'Items Held: 4' and a date range of '03/31/2025, 10:40:00 AM - 04/03/2025, 10:40:00 AM'. A red box highlights a right-pointing arrow button on the right side of this card. Below the active section is the 'Past Reservations' section, which contains a card for 'example reservation 2' with 'Items Held: 1' and a date range of '02/18/2025, 04:43:00 PM - 02/19/2025, 04:43:00 PM'. A right-pointing arrow button is visible on the right side of this card.

Creating Reservations

To create a reservation, click on the plus button in the top right corner of the reservations page (as shown in the previous section). Then, fill out all the necessary information. Please keep in mind that you need to select the check in and check out dates before you are able to select the equipment (red box). To add more equipment to the reservation, click on the plus button. To remove an item, click on the red minus button (shown in red box on the second picture).



The screenshot shows the 'Create a New Reservation' form. It includes the following fields: 'Reservation Name:' with a text input containing 'ex) Bartram Photoshoot...'; 'Choose Item(s):' with a dropdown menu showing 'Select...' and a plus button (+) on the right, which is highlighted by a red box; 'Client or Internal Team:' with a dropdown menu showing 'Select...' and a downward arrow; 'Check-Out Date:' with a date input field showing 'mm/dd/yyyy, --:-- --' and a calendar icon; 'Check-In Date:' with a date input field showing 'mm/dd/yyyy, --:-- --' and a calendar icon; and a blue 'Reserve' button at the bottom.

Create a New Reservation

Reservation Name:

Choose Item(s):

100ct AA Batteries

+

×

▼

Client or Internal Team:

Other

×

▼

Select...

▼

+

Enter Client/Team Name:

Check-Out Date:

04/16/2025, 03:29 PM

🗓

Check-In Date:

04/17/2025, 03:29 PM

🗓

Reserve

Checking Out/In Equipment

To get to the page to check out and in equipment, go to the reservations page and click on the arrow button, as explained in the beginning of the reservations section. The page will show your reservation information and list all the items in the reservation.

To check out an item, click on the “check out” button in the red box. This will cause a pop up to open up. Make sure that the scanner is connected to your device and turned on. In the pop up, press the “start scanning” button and scan the barcode on the equipment. Once the device beeps, it is scanned.

To check an item back in, do the same process under the “Check In Items” column.

To add an item to the reservation, click on the plus icon in the green box. It opens up a popup where you can choose an item to add. To remove an item from the reservation, click on the minus icon in the orange box. It opens a confirmation popup where you verify that you want to remove the item.

To extend the reservation, click on the button in the blue box. This will open up a popup where you can enter which date to extend to. Finally, to cancel the reservation, you can click on the button in the purple box. This will open up a confirmation popup to make sure that you want to cancel the reservation. You can only cancel the reservation if no items have been checked out.

test for user guide

04/15/2025, 03:32:00 PM - 04/19/2025, 03:32:00 PM

Team: Comms

Items Held: 2

Extend Reservation

Cancel Reservation

Check Out Items:

3 Pin XLR Male to Female #3:
ti38inbxWCzngKNnyk8

Check Out

Amaran Small LED #1:
Qk3SF7vVOWP54NHUfNij

Check Out

+

Check In Items:

Returned Items:

Check Out

1. Please connect the scanner to your device

2. Press the "Start Scanning" button

3. Scan the barcode on the item.

Start Scanning

Choose an item to add below:

Select...

Atomos Ninja V Monitor

KSIPZE LED Light Strip #1

Canon 90D Battery #1

Selens 5-in-1 Diffuser & Reflector

Lavalier Mic Kit #1

Canon C200

Hpusn Softbox Lighting Kit #1

The image displays three sequential screenshots of a web application's confirmation dialogs. Each dialog is a light gray box with a dark border and a close button (an 'x' in a circle) in the top right corner.

The first dialog asks: "Are you sure you want to remove 3 Pin XLR Male to Female #3 (ti38inlxxWCzngKNnyk8) from the reservation?". It features a blue "Confirm" button at the bottom center.

The second dialog asks: "Choose the new check-in date below:". It contains a text input field with the placeholder "mm/dd/yyyy, --:-- --" and a calendar icon on the right. A blue "Confirm" button is located at the bottom center.

The third dialog asks: "Are you sure you want to cancel the reservation?". It also has a blue "Confirm" button at the bottom center.

Overdue Equipment

If an item becomes overdue, an email will be sent to all admins and to the user who has the overdue item. The email will look like the following:

From: The Agency at UF <theagencyatufdevs@gmail.com>
Date: March 25, 2025 at 2:50:01 PM EDT
To: vallee.maeson@gmail.com
Subject: Overdue Reservation Alert: test reservation

The following reservation has overdue items:

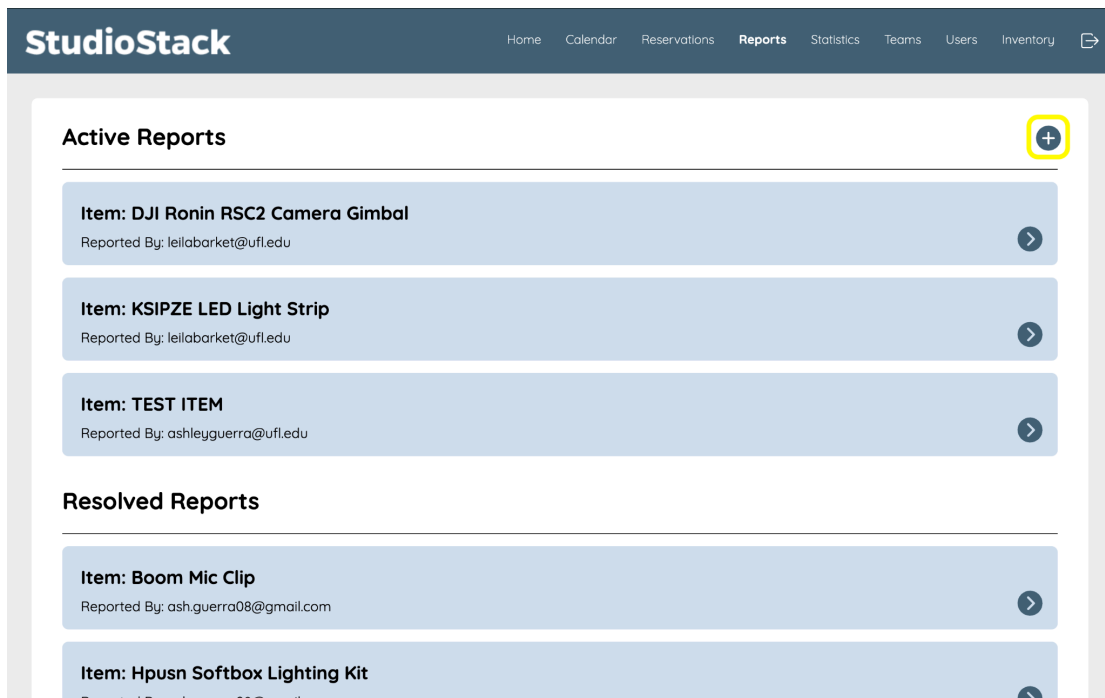
Reservation Name: test reservation

User: ash.guerra08@gmail.com

Items overdue: Black Backdrop Paper Roll (5PT7m0lfa5885mtkWXqo), SD Card - 128GB (9F1znxjL7Rym7Fr5D74Y), SD Card - 128GB (Bfi1mzJqfn43nXoYMNCf)

VI. Reports

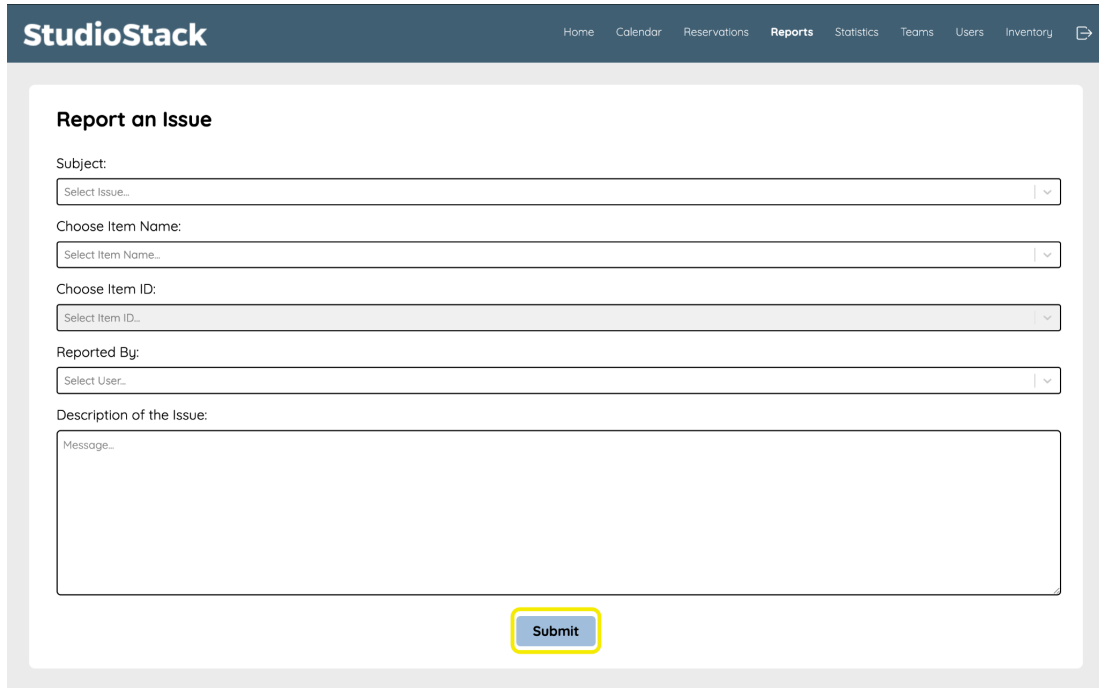
The reports page shows users their active and resolved reports. To create a new report, the user can click on the icon on the top right of the Reports page, which will direct them to a page where they can enter information about the report they are making.



Submitting Reports

To create a report, the user will click on the top right plus icon. Once they are on a page that looks like the page below, they can enter the category regarding the type of issue, the name of the item, the item ID, who it was reported by (their email), and a brief description of the issue or why they are reporting this item. Once all the information has

been filled out, they can click on the submit button. All information fields must be filled out or the user will receive an error.



The screenshot shows the 'Report an Issue' form in the StudioStack application. The form is located within a dark blue header bar that contains the StudioStack logo and a navigation menu with links to Home, Calendar, Reservations, Reports, Statistics, Teams, Users, and Inventory. The form itself is titled 'Report an Issue' and contains several input fields: 'Subject:' with a dropdown menu labeled 'Select Issue..', 'Choose Item Name:' with a dropdown menu labeled 'Select Item Name..', 'Choose Item ID:' with a dropdown menu labeled 'Select Item ID..', 'Reported By:' with a dropdown menu labeled 'Select User..', and 'Description of the Issue:' with a large text area labeled 'Message..'. A yellow 'Submit' button is located at the bottom right of the form.

Resolving Reports

To resolve a report, the admin can select any report listed under their “Active Reports”. This list will show all the reports that users have submitted, including any they have submitted themselves. On a non-admin user’s Reports page, they will only be able to see their own submitted reports under “Active Reports” and “Resolved Reports”. When the admin opens a report, they can view all the details that were submitted, such as the subject/category, item name and ID, who it was reported by, and a brief description. The admin can resolve this report by clicking the Resolve button at the bottom of the page. Then, a blue bar will appear at the top of the page indicating that the admin has resolved the report at that moment in time. The user will receive a notification on their dashboard indicating this as well.

StudioStack

[Home](#) [Calendar](#) [Reservations](#) [Reports](#) [Statistics](#) [Teams](#) [Users](#) [Inventory](#) [🔗](#)

Report Summary

Reported On: 04/07/2025, 02:56:28 PM

Subject:

App Not Working

Reported By:

leilabarket@ufl.edu

Item Name:

DJI Ronin RSC2 Camera Gimbal

Item ID:

BNxatUB2T5F4bNZWZXI6

Description of the Issue:

test

Resolve

StudioStack

[Home](#) [Calendar](#) [Reservations](#) [Reports](#) [Statistics](#) [Teams](#) [Users](#) [Inventory](#) [🔗](#)

Report Summary

Reported On: 04/07/2025, 02:56:28 PM

👍 This issue was resolved by ashleyguerra@ufl.edu on 04/19/2025, 07:49:28 PM.

Subject:

App Not Working

Reported By:

leilabarket@ufl.edu

Item Name:

DJI Ronin RSC2 Camera Gimbal

Item ID:

BNxatUB2T5F4bNZWZXI6

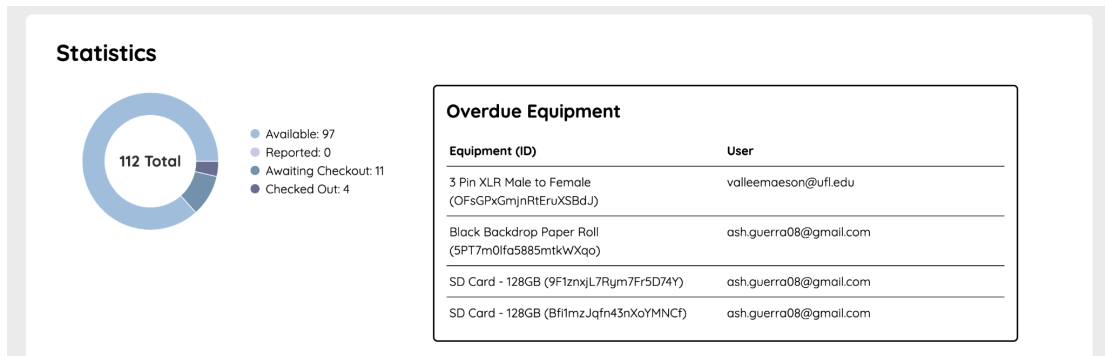
Description of the Issue:

test

VII. Statistics

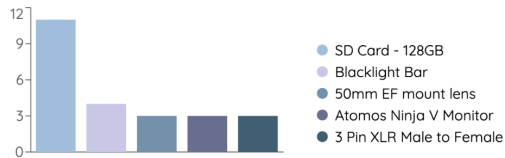
The statistics page shows key information about the website. This page is only viewable by admins. The pie chart shows the statuses of the equipment. There is also an overdue box that shows which equipment is currently overdue. These are both in the first picture

below. The second picture shows one of the sections of graphs, with this one being the reservations data. You can click on the buttons in the red box to change the time frame for the graph. Finally, in the third picture, it shows the record of overdue equipment, by showing which users have had overdue equipment.



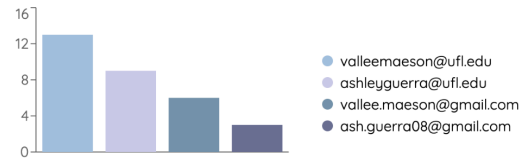
Reservations Data

Top Reserved Items: All-Time



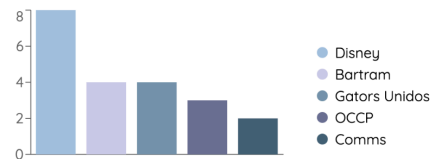
Past 30 Days Past 60 Days Past 90 Days All-Time

Top Reserving Users: All-Time



Past 30 Days Past 60 Days Past 90 Days All-Time

Top Teams: All-Time



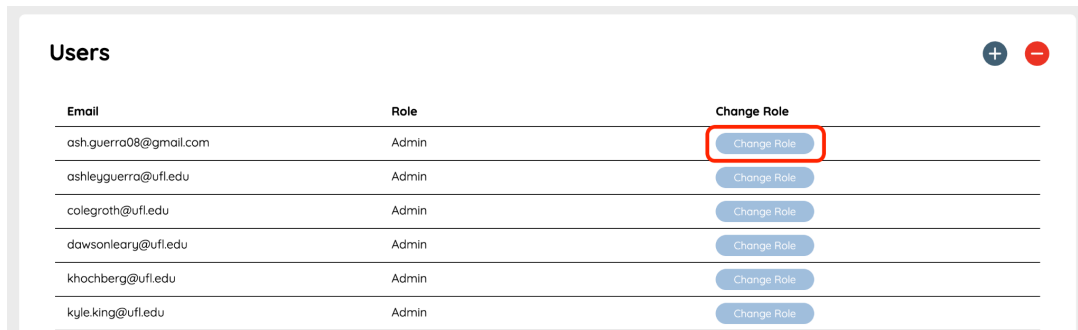
Past 30 Days Past 60 Days Past 90 Days All-Time

Overdue Equipment Record

User	Total Instances	Average Overdue Time	Overdue Times
valleemaeson@ufl.edu	3	13d 15h 25m	25d 23h 8m, 1d 7h 42m
ash.guerra08@gmail.com	3	N/A	

VIII. Users

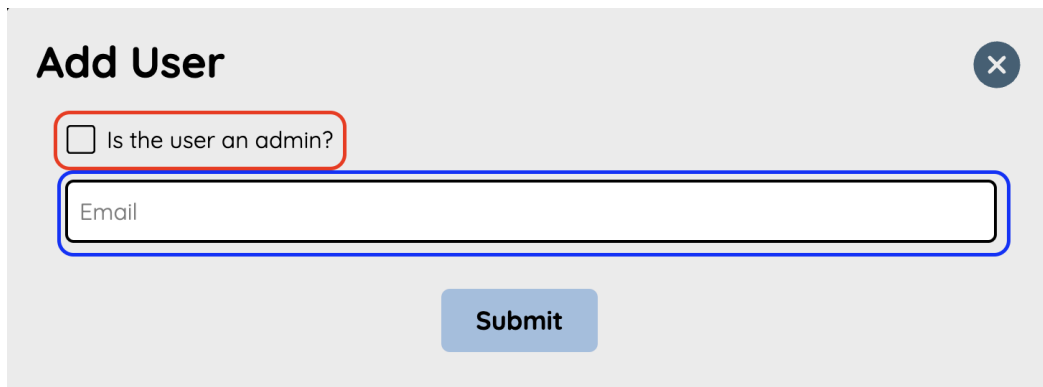
The users page allows admins to add and remove users. Only admins can see this page, and it consists of a list of users with an indication of their role. You can click on the plus button in the top right corner to add a user and the minus button will remove a user. Plus, by clicking the “Change Role” button shown in the red box, you can change the role of a user from a student to an admin and vice versa.

The screenshot shows a web interface titled "Users". In the top right corner, there are two circular buttons: a blue one with a white plus sign and a red one with a white minus sign. Below the title is a table with three columns: "Email", "Role", and "Change Role". The table contains six rows of user data. In the first row, the "Change Role" button is highlighted with a red rectangular box.

Email	Role	Change Role
ash.guerra08@gmail.com	Admin	Change Role
ashleyguerra@ufl.edu	Admin	Change Role
colegroth@ufl.edu	Admin	Change Role
dawsonleary@ufl.edu	Admin	Change Role
khochberg@ufl.edu	Admin	Change Role
kyle.king@ufl.edu	Admin	Change Role

Adding Users

After clicking on the plus button, a popup will appear. You can add a user by specifying the role, as shown in the red box, and by entering the email address, as shown in the blue box. When ready, click submit.

The screenshot shows a "Add User" popup window. It has a title bar with a close button (an 'x' in a circle). The form contains a checkbox labeled "Is the user an admin?" which is highlighted with a red rectangular box. Below the checkbox is a text input field with the placeholder text "Email", which is highlighted with a blue rectangular box. At the bottom of the form is a blue "Submit" button.

Add User

☐ Is the user an admin?

Email

Submit

Removing Users

After clicking on the minus button, a popup will appear. You can remove a user by going through the dropdown and clicking on a user you would like to remove. When ready, click submit.

Remove User

Select...

ash.guerra08@gmail.com

ashleyguerra@ufl.edu

theagencyatufdevs@gmail.com

vallee.maeson@gmail.com

valleemaeson@ufl.edu

IX. Teams

The teams page allows users to view all of the Internal and Client teams that the Production Department is working on. This page functions similarly to the Users and Inventory pages.

StudioStack

Home

Calendar

Reservations

Reports

Statistics

Teams

Users

Inventory

Teams

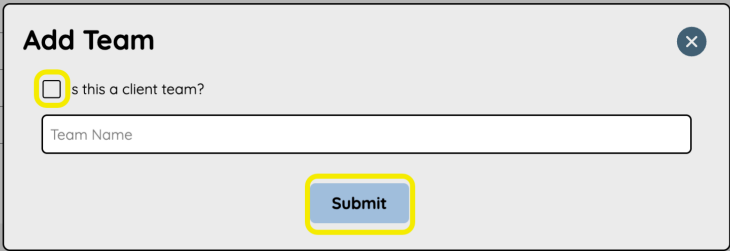
+

-

Name	Type
Bartram	Client
Comms	Internal
Disney	Client
Gators Unidos	Client
OCCP	Client

Add Team

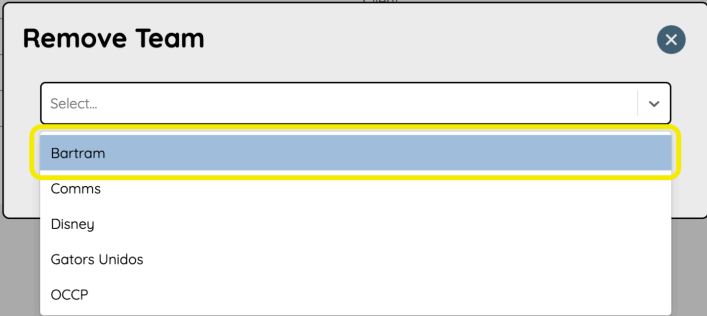
To add a team, the user can select the plus icon on the top right. A popup will appear asking the user if they are adding a client team. The user can select this to indicate “yes”, and leave it unchecked to indicate “no”.



The image shows a modal window titled "Add Team" with a close button (X) in the top right corner. Inside the modal, there is a checkbox labeled "Is this a client team?". The checkbox is currently unchecked. Below the checkbox is a text input field labeled "Team Name". At the bottom of the modal is a blue "Submit" button. The entire modal is set against a background of a table with a header row labeled "Type".

Remove Team

To remove a team, the user can select the minus icon on the top right. A popup will appear with a dropdown containing all of the team names. Then, the user can select the team they would like to remove. Once it has been selected, the user can click submit.



The image shows a modal window titled "Remove Team" with a close button (X) in the top right corner. Inside the modal, there is a dropdown menu with the text "Select..." and a downward arrow. The dropdown is open, showing a list of team names: "Bartram", "Comms", "Disney", "Gators Unidos", and "OCCP". The "Bartram" option is highlighted with a blue background. Below the dropdown is a blue "Submit" button. The entire modal is set against a background of a table with a header row labeled "Type" and a second row labeled "Client".

X. Inventory

The inventory page allows admins to add and remove equipment. Only admins can see this page, and it consists of a list of equipment showing their category, status, and barcode number. You can click on the plus button in the top right corner to add an item and the minus button will remove an item (as shown in the red box). You can also sort the inventory items by category clicking the respective category in the orange box. Plus, you can search through the inventory list by name with the search bar shown in the purple box. The item's status is denoted by an exclamation mark (meaning it is reported) or a check mark (meaning it is not broken), as shown in the green box. You can also download an item's barcode by clicking on the button in the blue box.

Item Name	Category	Status	Download Barcode
TEST ITEM	Miscellaneous	!	lcD7V7NGSdsin8mlvuH
3 Pin XLR Female to 3.5mm Adaptor	Audio	✓	zTMVtUcveh3icDNs2jgkl
Power Strip	Cables & Cords	✓	zh4G4R0lbTO1VOdmqeOh
50mm EF mount lens	Camera & Accessories	✓	xwztVouQmjgk3IfkGO23
Sand Bag (Small) #2	Miscellaneous	✓	wpzbS3ceZSiwNxiAtp
White Balance Posterboard	Camera & Accessories	✓	vkGNCfuJiffxcvG6gkT
Emart Backdrop Kit #2	Lights	✓	u5i:BN55hzf7wvnt44KTh
Neewer Speedlite	Lights	✓	u6artUGV8m9QJVq7INiL

Adding Equipment

After clicking on the plus button, a popup will appear. You can add an item by entering the item name and by selecting the category.. When ready, click submit.

The 'Add Item' popup form is a light gray box with a close button (X) in the top right corner. It contains two input fields: 'Enter Item Name' and 'Select Category'. The 'Select Category' dropdown menu is open, showing a list of categories: 'Camera & Accessories', 'Audio', 'Lights' (highlighted in blue), 'Production Design', 'Cables & Cords', and 'Miscellaneous'.

Removing Equipment

After clicking on the minus button, a popup will appear. You can remove an item by going through the dropdown and clicking on an item name you would like to remove (as shown in the picture). When ready, click submit.

The 'Remove Item' popup form is a light gray box with a close button (X) in the top right corner. It contains a dropdown menu labeled 'Select...'. The dropdown menu is open, showing a list of items: 'TEST ITEM', '3 Pin XLR Female to 3.5mm Adaptor', 'Power Strip', '50mm EF mount lens', 'Sand Bag (Small)', 'White Balance Posterboard' (highlighted in blue), 'Emart Backdrop Kit', and 'Cables & Cords'.

XI. Miscellaneous

1. If an item becomes overdue, an email will be automatically sent to every admin and to the user who has the overdue item. This email will state which item and which reservation this belongs to.
2. If you have an overdue item, a notification will also appear at the top of the reservations page.
3. If an item is reported and a reservation in the future has this item reserved, an email will be sent to the user of the reservation. This user will have the ability to change out the piece of equipment to a non-reported one. Please see the section for adding items to a reservation.

XII. FAQ

Q: Why can't I login to StudioStack?

A: You might not be an approved user. Please check with your Department Manager to see if you are added as a user.

Transition Plan/Next Steps

The next step for StudioStack is to transfer the codebase to the Development Department at The Agency. We have spoken with our advisor, Natalie Asorey, and have received approval to do so. During the transition process, we will assist the development team with forking the repository from Maeson's Github, which is where StudioStack is currently being stored. We will provide all necessary documentation, including a user guide, a backlog of planned features and/or stretch goals, and important credentials. Our long-term goals for this project are to enhance the user-experience, fix bugs, and introduce new features/iterations as the production department uses the website and provides feedback. We hope that the Development Department at The Agency can continue to collaborate with the Production Department to improve and grow StudioStack beyond the end of this semester.

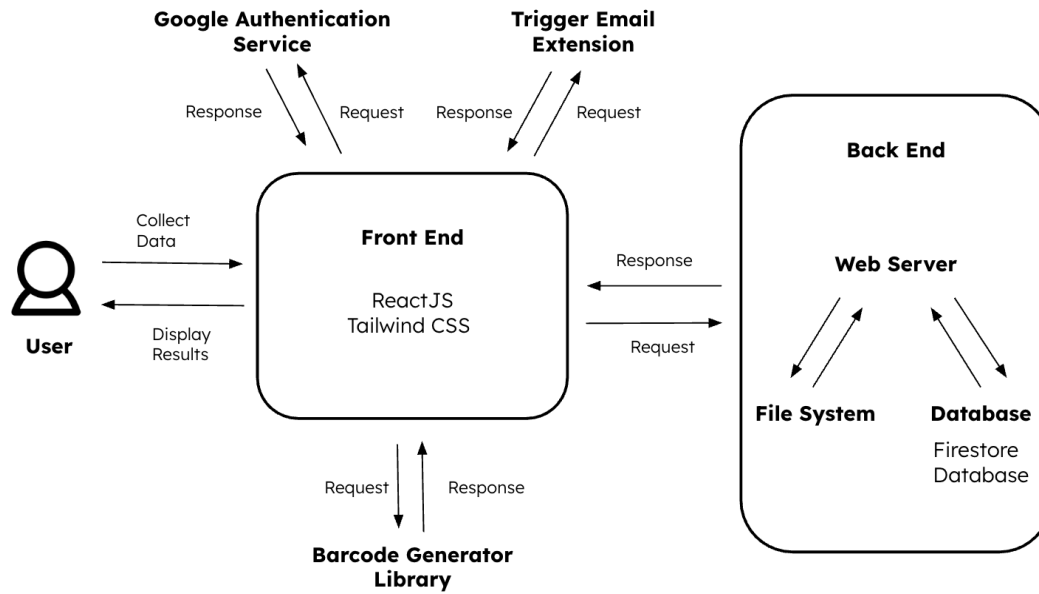
Acknowledgements

We give a special thanks to everyone who helped out with the making of StudioStack. These amazing people include:

- Natalie Asorey, Faculty Advisor & Associate Director at The Agency at UF
- Dawson Leary, Production Department Manager at The Agency at UF
- Mark Rottensteiner, Accounts Director at The Agency at UF
- The Production Department at The Agency at UF

Appendices

Figure 1: Software Architecture Model



Biography

Ashley Guerra

I am a Senior majoring in Computer Science and minoring in Mass Communications at the University of Florida. Over the course of my undergraduate career, I have taken on several roles, such as a UX Designer, Project Manager, IOS App Developer, Full Stack Developer, and Developer Department Manager. I have been primarily involved with organizations such as Gator User Design, the Society of Hispanic Professional Engineers, and The Agency at UF. In summer of 2024, I worked at Bank of America as a Software Engineer intern, where I will be returning for full-time employment. During my free time, I love playing guitar, going to workout classes, wheel throwing at a ceramics studio, and going to concerts. I aspire to work in product design or user experience design in the future!

Maeson Vallee

I am a Senior Computer Science student at the University of Florida. Since the summer of 2022, I have worked at The Agency at UF, working as a Data Analyst then a Full-Stack Developer. For the 2023-2024 school year, I worked as Development Department Manager for The Agency at UF. I have worked as a TA for Programming Fundamentals II for the Spring 2023, Fall 2023, and Spring 2024 semesters. In the summer of 2023, I worked as a Software Development Intern for Texas Instruments. In the summer of 2024, I interned at Capital One as a Software Engineer Intern, working on their IOS mobile application. I like reading books in the sun, and enjoy going out to try new restaurants.